1.4: Composition of Functions

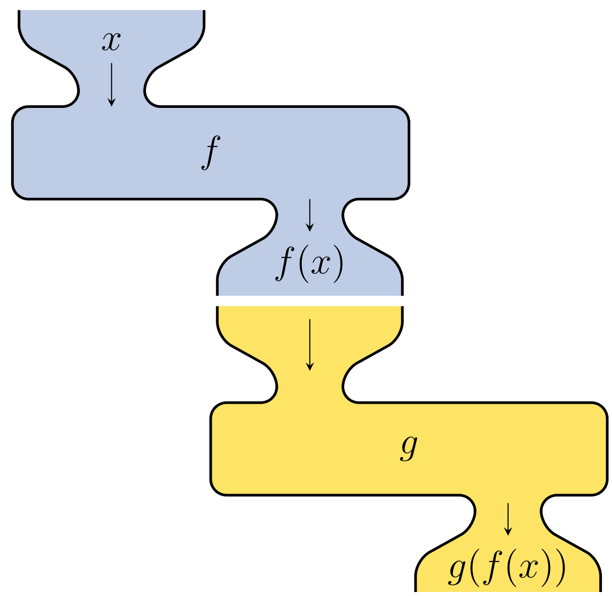
Definition. (The Composition of Two Functions)
Let f and g be functions. Then the composition of g and f is the function $g \circ f$ defined by

$$(g \circ f)(x) = g(f(x))$$

The domain of $g \circ f$ is the set of all x is the domain of f such that $f(x)$ lies in the domain of g.

Example. Let $f(x) = \sqrt{x+1}$ and $g(x) = 4 - x$. Find the domain of the following:

$g(f(x)) =$



$f(g(x)) =$

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Example. For the following, find functions $g(x)$ and $h(x)$ so that $f(x) = g(h(x))$:

$$f(x) = \frac{1}{x+2}$$

$$f(x) = \sqrt{x-4}$$

$$f(x) = \left(x^2 + 3x - 8\right)^4$$

$$f(x) = \frac{x-3}{\sqrt{-(x-3)}}$$